Cutnell & Johnson
Newton's Second Law Part 2
Constant Force
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

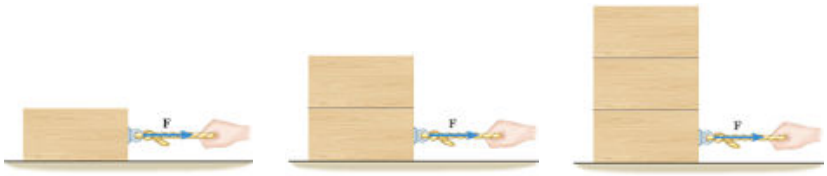
Date:	
Your Name:	
Group:	
Period:	

Introduction *Exploration Goals*

§ The most important equation in physics is F=ma, also known as Newton's Second Law of mechanics. It governs the behavior of everything that is seen and unseen on Earth and in the cosmos -- from the trajectory of a baseball to the motion of a planet.

§ The purpose of this exploration is to find out what happens to an object's acceleration when the net force applied to the object is constant and the object's mass is changed.

§ Use a Motion Sensor to measure the motion of an object that is accelerated by a net force. Determine what happens to the acceleration of the cart when the net force is constant and the total mass is increased.



Note: In Newton's Second Law Part 1 you measured the acceleration of an object when the net force increases but the total mass is constant.

Learning Outcomes

You will be able to:

- Interpret data from a graph of velocity versus time to find acceleration.
- Measure the acceleration of an object when a constant net force is applied and the mass is increased.
- Calculate theoretical values for acceleration based on net force and mass and compare the theoretical values with the measured values.
- Describe the relationship between the acceleration of an object, the net force applied to the object, and the mass of the object.

Hypothesis

§ What happens to an object when you apply a net force to it?

§ What happens to the moton of an object if you change the mass of the object but keep the net force constant?

Background

§ Newton's first law states that if no net force acts on an object, then the velocity of the object remains unchanged. The second law deals with what happens when a net force does act.

§ As long as a net force acts, the velocity of an object changes - in other words, it accelerates. If more force is applied, the greater force produces a greater acceleration.

§ In Newton's second law, the net force is only one of two factors that determine the acceleration. The other is the inertia or mass of the object. Newton's second law states that for a given net force, the magnitude of the acceleration is inversely proportional to the mass. Twice the mass means one-half the acceleration, if the same net force acts on the object.

§ Thus, the second law shows how the acceleration depends on both the net force and the mass.

$$a \approx F_{net}$$

» Acceleration is directly proportional to net force.

$$a \approx \frac{1}{m}$$

» Acceleration is inversely proportional to mass.

$$a = \frac{F_{net}}{m}$$

» Newton's Second Law of Motion

Materials

- PASPORT USB Link
- PASPORT Motion Sensor
- Mass and Hanger Set
- IDS 1.2 m Track w/ Carts
- Balance
- String, 2m
- Super Pulley with Clamp

Setup *Equipment Setup*

Safety Reminder: Follow the directions for using the equipment.

» You can use the same equipment setup that you used in Part 1 of Newton's Second Law.

§ The Dynamics Track has a fixed end stop at one end. Place the Motion Sensor on the track in front of the fixed end stop.

§ Attach the Pulley to the other end of the Dynamics Track. If you have a piece of wood you can place this in front of the pulley to stop the cart.

§ Level the track. Put the cart on the track. If the cart moves one way or the other, use the leveling screw under the fixed end stop to level the track.

§ Attach a piece of string about 1.2 m long to one end of the cart.

Note: It is very important that the track is level to get the best readings.

Setup *Equipment Setup*

§ Add a 20-g mass to the mass hanger and weigh the hanger plus the mass. Record the mass of the hanger plus the one 20-g mass.

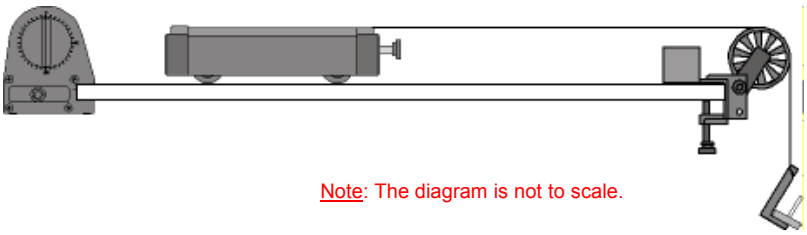
§ Weigh the cart and record the initial mass of the cart.

§ Attach the mass hanger to the other end of the string, and put the string in the pulley's groove.

§ Adjust the pulley up or down so the string is parallel to the track as shown in the diagram.

» Run #1: Total mass of the hanger and 20-g mass (mH):

» Run #1: Total mass of the cart (mC):



Note: The diagram is not to scale.

Process *Prepare to Record Data: Constant Net Force*

Read this page before you take data. The next page is where you will record data.

§ You will take three runs of data. For Run #1, use the arrangement described on the previous page (one 20-g mass added to the mass hanger).

» For the second and third runs, add mass to the tray on the cart.

§ For Run #2, add one 100-g mass to the tray of the cart before you record data. The total mass of the system is increased, but the net force is the same.

§ For Run #3, add a second 100-g mass to the tray of the cart before you record data. As in Run #3, the total mass of the system is increased, but the net force is kept constant.

Note: Catch the cart before it hits the pulley or you could damage the pulley.

Process *Record Data: Constant Net Force, Increasing Mass*

§ Click **Start**. Release the cart. Stop the cart before it hits the pulley and click **Stop**.

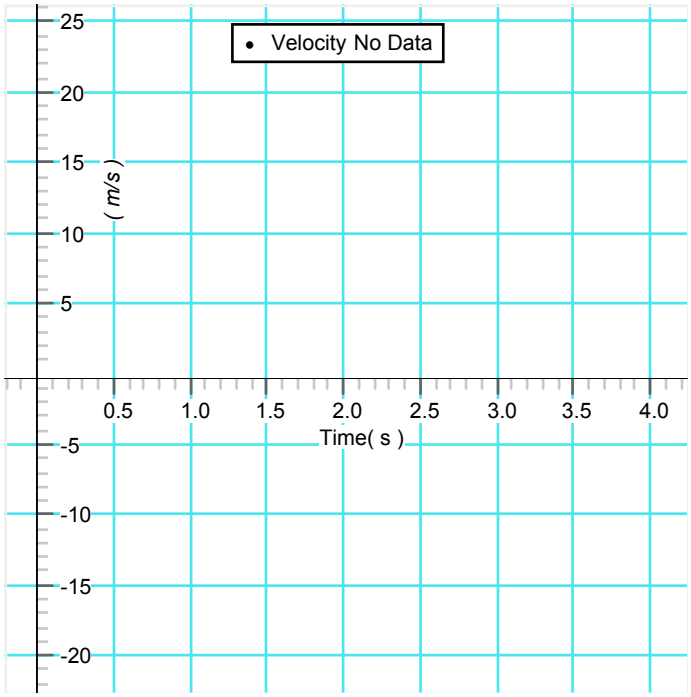
» The next page describes how to determine the acceleration.

§ For Run #2, add a 100-g mass to the cart. Record the total mass of the cart plus the extra mass. Record data as before.

§ For Run #3, add the second 100-g mass to the cart. Record the total mass of the cart plus the extra masses. Record data as before.

» Run #2: Total mass of cart plus mass

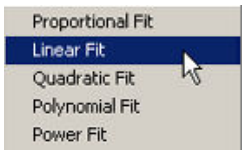
» Run #3: Total mass of cart plus masses



Analyze *How to Find the Acceleration from the Velocity*

§ Click **Scale to Fit** to rescale the graph if needed. Use the cursor to highlight a smooth part of the graph.

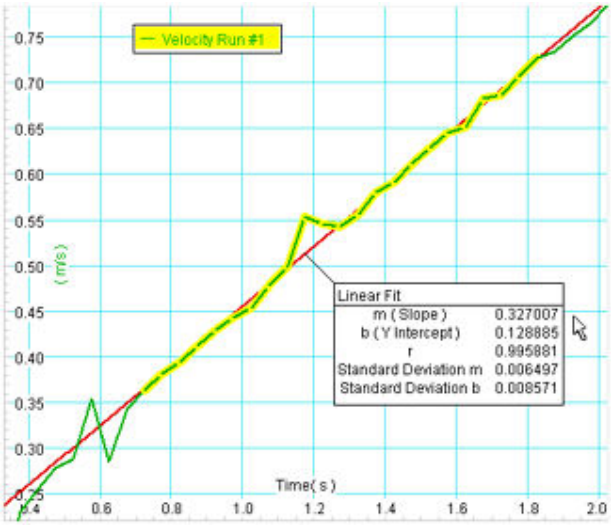
§ Click the **Fit** menu and choose **Linear Fit**.



§ The slope "m" of the best fit line is the acceleration.

§ After recording the slope, use the Data menu to select the next run.

» Analyze your data on the next page.



This example shows the highlighted area and the Linear Fit information box. Note that the Scale to Fit button, Fit menu, and Data menu are not shown.

Analyze *Determine the Acceleration from the Velocity*

§ Select **Run #1**. Click **Scale to Fit** if needed. Highlight part of the graph that shows increasing velocity. Select **Linear Fit** from the **Fit** menu.

§ Record the value of "m" (slope) as the acceleration for Run #1.

§ After recording the slope, use the **Data** menu to select the next run.

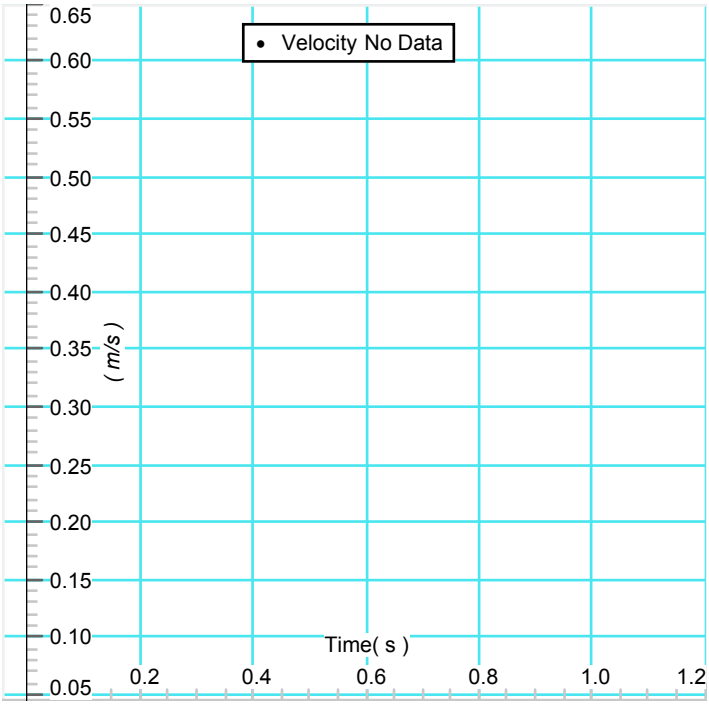
§ Repeat the process for Run #2 and Run #3.

» The slope "m" of the best fit line is the acceleration.

Run #1 acceleration

Run #2 acceleration

Run #3 acceleration



Observations

§ What did you observe about the slope of the Linear Fit as the total mass was increased but the net force was kept constant? Why did the slope change for each run?

§ What variable(s) did you keep the same from one run to the next? What variable(s) did you control?

§ What variable changed as a result and what was the response?

Analyze *Calculations: Constant Net Force, Increasing Mass*

§ Calculate the theoretical acceleration when mass is increased and net force is constant.

§ The acceleration is the ratio of the net force divided by the total mass. The total mass of the system (mass of cart plus mass of hanger) increased, but the net force (mass of hanger x 9.8) was constant for each run.

§ Assuming no friction, the net force is the weight of the hanger (mass x 9.8).

$$a = \frac{m_H g}{m_c + m_H}$$

g = 9.8 m/s/s

	Total mass of the cart (m_C)	Total mass of system ($m_C + m_H$)	Acceleration (theoretical)
Run #1:			
Run #2:			
Run #3:			

Total mass of hanger and one 20-g mass* (m_H): _____

Net force* (m_H x 9.8) _____

*Remains constant.

Analyze *Comparison*

§ Compare the measured acceleration to the theoretical acceleration for each run. Find the percent difference (% Difference).

	Theoretical	Measured	% Difference
Run #1:			
Run #2:			
Run #3:			

$$\% \text{ difference} = \left| \frac{\text{measured} - \text{theoretical}}{\text{theoretical}} \right| \times 100$$

§ How do your results compare to your hypothesis?

Hypothesis

Synthesize *Conclusions*

§ What happens to an object's acceleration if the net force applied to the object is constant but the total mass increases?

§ Is the object's acceleration inversely proportional to the total mass?

§ What were the sources of error in this experiment?

Synthesize *Applications and Extension Problem*

§ Based on Newton's Second Law, predict what will happen to a truck's acceleration if it gains mass during a rainstorm but the net force on the truck remains constant.

§ An unloaded sled with a mass of 12 kg has an acceleration of 2.3 m/s/s when it is pushed on a horizontal surface with a net force of 27.6 N. What is its acceleration when it is fully loaded to 25 kg and the net force is the same?